FOOD SAFETY × IoT × EMBEDDED SYSTEMS

IoT-Based Cooking Oil Adulteration Detector

Check the oil. Read the result. Share it instantly.

ESP32 | Arduino C/C++ | MQTT | Node-RED | Phase 1: Design & Literature Survey | 2026-27

A low-cost, portable, reagent-free screening device that flags possibly adulterated cooking oil using optical sensing, an ESP32 and a live IoT dashboard.

The Problem

Cooking oil is in almost every kitchen, yet it is often mixed with cheaper oils such as **palm oil, mineral oil or used (recycled) oil** to increase profit. In some cases, toxic substances like **argemone oil** are added, which is a serious risk to public health.

Laboratory methods such as **gas chromatography and FTIR** can detect adulteration accurately, but they are **slow, costly and need trained people**. That means they cannot be used at a shop counter or in a home kitchen, which is exactly where a quick check is needed.

> **Note**
> **Problem statement:** There is a need for a low-cost IoT-based system that can detect possible cooking oil adulteration quickly and provide the results through a digital platform.

Our Solution

A compact detector that checks a **fixed volume of oil** using a **turbidity sensor** and an **IR sensor**. An **ESP32** compares the readings with the stored values of pure oil, shows the verdict on an **OLED display**, and publishes the data over **Wi-Fi (MQTT)** to a **Node-RED dashboard**.

Low cost
Built from off-the-shelf modules

Portable
Small enough to use at the point of sale or use

Reagent-free
No chemicals needed, only light passing through the oil

Instant result
Shown on the OLED right at the place of testing

Two optical sensors
Turbidity (light scattering) plus IR absorbance for a more reliable decision

Graded verdict
Pure, Borderline, Possibly Adulterated or Highly Suspected, not just yes/no

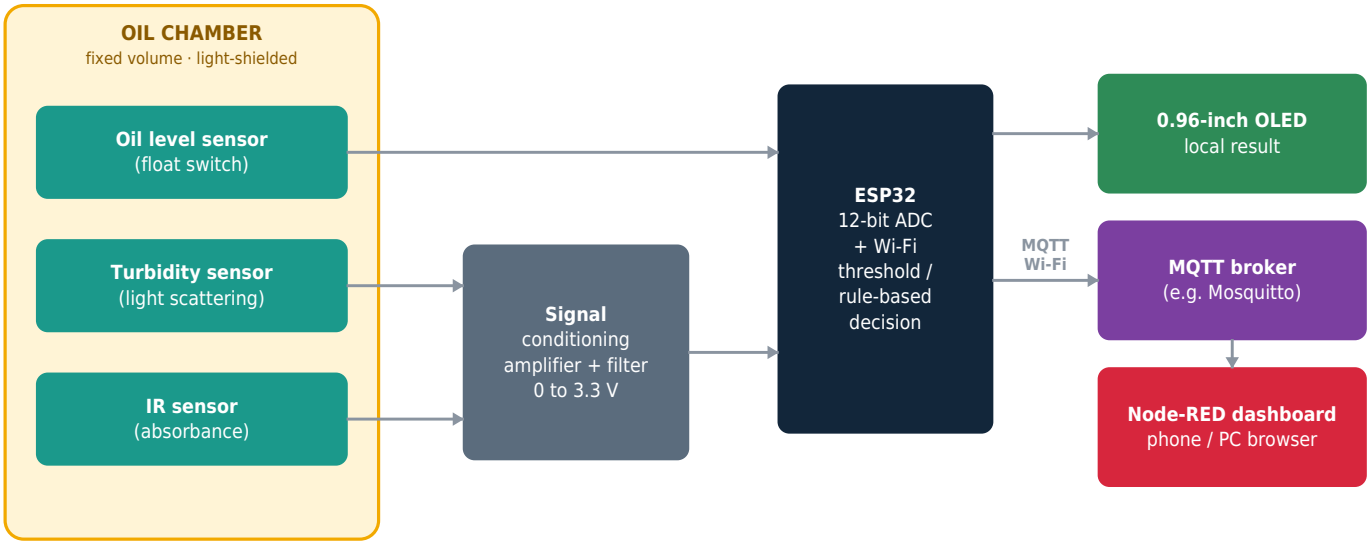
Connected
MQTT + Node-RED give history, gauge and alerts in any phone or PC browser

Lab-aware
Positive samples are sent to an accredited laboratory for confirmation

> **Important**
> This device is a **first-screening tool**. It does not replace laboratory analysis. Any sample flagged as suspicious should be **confirmed in an accredited laboratory**.

How It Works

Adulteration changes how an oil scatters and absorbs light. The detector turns that change into numbers.



System block diagram

Step by step

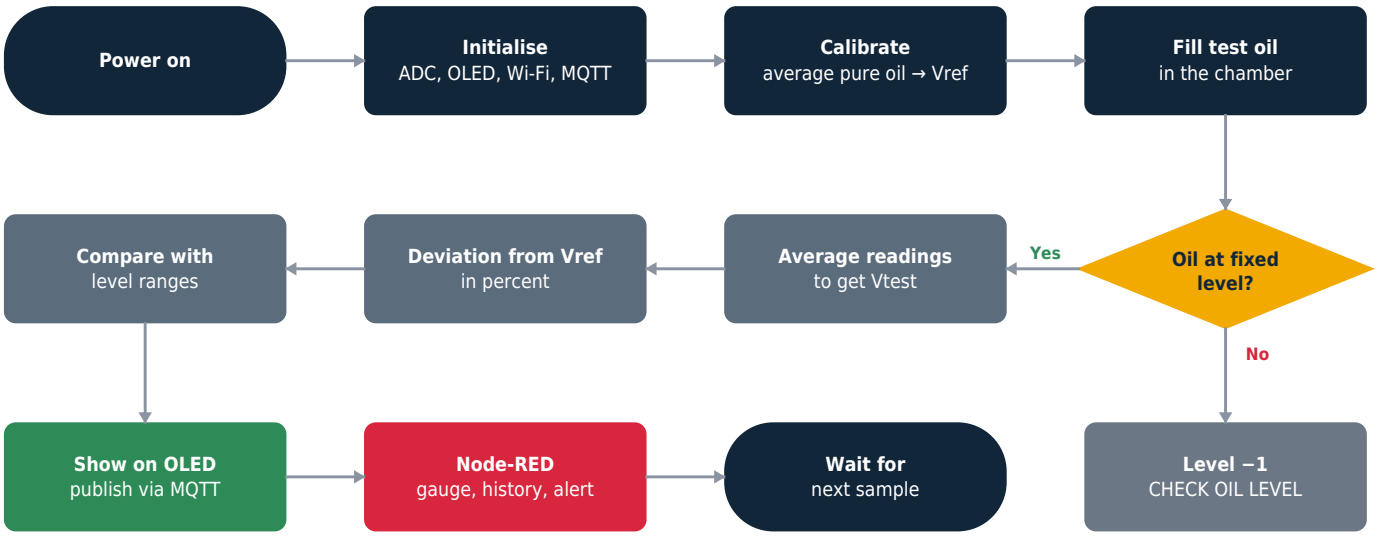
1. **Fill and verify.** A fixed volume of oil goes into the chamber. The oil level sensor (float switch) confirms the level is correct before any measurement is made.
2. **Sense.** The turbidity sensor measures light scattering, and the IR sensor measures infrared absorbance.
3. **Condition.** An amplifier and filter clean up the small, noisy sensor signals and bring them into the ESP32's 0–3.3 V ADC range.
4. **Compare.** The ESP32 averages several ADC readings and compares them with the stored **pure-oil reference** of the same oil type.
5. **Decide.** A threshold / rule-based algorithm classifies the oil into a result level.
6. **Show and share.** The verdict appears on the OLED, and the data is published over MQTT to the Node-RED dashboard.
7. **Confirm.** Suspected samples go to an accredited laboratory.

The decision rule

```
Deviation (%) = |Vtest − Vref| / Vref × 100
```

`Vref` is the averaged reading of pure oil of the same type, and `Vtest` is the averaged reading of the oil being tested.

Firmware flow



Level ranges: below 5% = Level 0, 5% to below 10% = Level 1, 10% to below 25% = Level 2, 25% or more = Level 3. An invalid reading (Level −1) is also published, then the user refills and retests.

Hardware

#	Component	Purpose
1	**ESP32 development board**	Main controller with ADC and Wi-Fi
2	**Analog turbidity sensor module**	Measures light transmission (turbidity) of the oil
3	**IR sensor module**	Measures infrared absorbance of the oil
4	**Signal conditioning circuit**	Amplifier and filter for the sensor signals
5	**Oil chamber and level sensor**	Holds a fixed volume of oil and detects its level
6	**0.96-inch I2C OLED display (SSD1306)**	Local result display
7	**Regulated 5 V power supply**	Powers the ESP32, sensors, signal conditioning and display
8	**MQTT broker (e.g. Mosquitto) with Node-RED**	Receives data; monitoring, history and alerts

Why this design?

- **Two sensors, one decision.** Turbidity responds to the clarity and colour of the oil, while IR responds to its infrared absorption, the same principle used in laboratory infrared spectroscopy. Two independent readings are more reliable than one.
- **Repeatable measurements.** A fixed volume, a fixed sensor position and a light shield mean the reading depends only on the oil.
- **Stable readings.** Amplification, filtering and averaging turn small, noisy signals into a steady value. A regulated supply keeps the light-source brightness constant, so variation is not mistaken for adulteration.
- **ESP32 inside.** The built-in ADC and Wi-Fi remove the need for an extra communication module, keeping the device compact and cheap.
- **MQTT + Node-RED.** MQTT is light-weight and well suited to small IoT devices, and Node-RED provides a ready dashboard with a gauge, history chart and alerts. Every reading is logged with its result for traceability.
- **I2C OLED.** It needs only two communication lines, which simplifies wiring.

Firmware: embedded C/C++ using the Arduino framework, covering calibration, measurement, decision making, display and MQTT communication.

Output Levels

Instead of a single yes/no, the result is graded. A **Borderline** level asks for a retest instead of raising a false alarm or missing a doubtful sample.

Level	Deviation from pure oil	OLED shows	MQTT status	Node-RED dashboard
0	below 5%	**PURE**	`PURE`	Gauge in green zone; value and history updated; no alert
1	5% to below 10%	**BORDERLINE – RETEST**	`BORDERLINE`	Gauge in amber zone; message “Repeat the test”; logged in history
2	10% to below 25%	**POSSIBLY ADULTERATED**	`POSSIBLY_ADULTERATED`	Gauge in red zone; red alert; sample to be sent to laboratory
3	25% or more	**HIGHLY SUSPECTED**	`HIGHLY_SUSPECTED`	Gauge at red end; strong red alert; sample to be sent to laboratory
−1	Oil level not at fixed mark, or ADC value out of range	**CHECK OIL LEVEL**	`INVALID`	Grey indicator “No valid reading”; not added to the history chart

> **Note**
> The percentage is the **deviation of the sensor voltage from Vref**. It is **not** the concentration of the adulterant. The ranges above are **initial values** and will be finalised from measured readings of pure and blended oil samples.

Example MQTT message

Published on the topic `oildetector/result`:

```
{
  "vtest": 1.82,
  "vref": 2.08,
  "deviation": 12.5,
  "level": 2,
  "status": "POSSIBLY_ADULTERATED"
}
```

IoT Dashboard

The ESP32 publishes each reading over Wi-Fi to the MQTT broker. **Node-RED** subscribes to the topic and shows:

- A **gauge** scaled from 0 to 50% deviation, with green (0–5%), amber (5–10%) and red (10–50%) zones
- A **history chart** of past readings
- An **alert message** for Level 2 and Level 3 samples

Because the interface opens in a **phone or PC browser**, inspectors and quality-control staff can check results from anywhere.

Who Is It For?

Audience	How they benefit
Food safety inspectors	Quick on-the-spot screening, with only suspicious samples sent for lab tests
Quality control staff	Digital record of every reading, with history and alerts
Consumers	A simple, reagent-free way to get an early warning about oil quality

Design Target

The prototype is designed to separate **pure oil** from oil adulterated at **10% or higher concentration**, using **sunflower oil and groundnut oil** samples. It needs **no chemical reagents** and gives stable readings that change with the quality of the oil.

Roadmap

- ☑ Problem identification and literature survey
- ☑ System design: block diagram, working principle and decision algorithm
- ☑ Output levels defined for OLED, MQTT and Node-RED
- ☐ Prototype testing and sensor calibration
- ☐ Measure readings of pure and blended samples, then finalise the deviation thresholds
- ☐ Validate separation of pure oil from oil adulterated at 10% or higher
- ☐ Future work: train a machine-learning classifier on the readings stored by the Node-RED / MQTT side

Research Background

The project is backed by a survey of **twenty studies**. Key takeaways:

- **Adulteration is long-standing, growing and increasingly sophisticated.** Mixing with cheaper oils is common, and ordinary quality checks often miss modern methods.
- **Accurate laboratory methods exist but are lab-bound.** HPLC detected palmolein and palm kernel oil in coconut oil at a minimum of 2–4% with 78–98% accuracy, and GC-MS with FTIR could signal pork fat in ghee at 1%. Both approaches need laboratory equipment and trained people.
- **Optical sensing works without reagents.** Fibre-optic sensors by Libish et al. detected paraffin oil in coconut and sunflower oil at 3–4% and adulteration of virgin olive oil at 4%, which supports light-based screening.
- **Quick, non-destructive screening is the way forward.** Reviews describe AI-based approaches as quick, non-destructive and scalable, compared with lengthy and destructive conventional techniques.
- **The gap this project fills:** a simple, low-cost, portable device that measures a physical property of the oil with off-the-shelf sensors, gives an on-the-spot result and sends the readings to a dashboard.

Selected references

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[5] A. H. Ali, A. Elbarbary, S. Q. Liu, M. Hachem, J. A. Mazumder, F. Banat, X. Wang and M. Ayyash, “Assessment of buffalo and bovine ghee authenticity using fatty acid profile analysis and FTIR spectroscopy,” *npj Science of Food*, vol. 10, art. 16, 2026. doi: 10.1038/s41538-025-00664-9.

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